SPY Tail Risk Analysis Report

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1. Introduction

The SPY ETF plays a central role in the U.S. equity market, serving as the primary benchmark for broad market exposure and a foundation for portfolio construction and risk modeling. Because of this, many practitioners rely on Gaussian assumptions and Sharpe ratio-based metrics when evaluating its behavior or projecting future performance. While convenient, these tools often fail to capture the magnitude of heavy downturn events such as the 2008 financial crisis or the 2020 COVID-19 pandemic. This analysis tests how well those assumptions align with SPY’s actual return distribution and assesses whether they adequately reflect the tail risks present in real-world data.

2. Data & Methodology

This analysis uses daily adjusted closing prices for the SPDR S&P 500 ETF (SPY) over the period September 21, 2017 to February 13, 2026. Data was pulled from Tradingview. Returns were computed as daily log returns. The dataset was cleaned to remove missing values and ensure consistent time indexing. The analysis includes descriptive statistics, a histogram of daily returns, a tail-risk assessment using the lower 5% of the return distribution, and a dynamic Sharpe ratio computed over a rolling 252-day window. All computations were performed in Python using pandas, NumPy, and Matplotlib.

3. Key Findings

- SPY's daily returns exhibit a mild negative skew (~-.3) and high excess kurtosis (~13.6), indicating a leptokurtic return distribution with heavier tails than a Gaussian model would assume.
- The lower 5% of returns show meaningfully larger losses than the normal distribution implies, with empirical tail risk significantly exceeding Gaussian estimates.
- During the worst 1% of days, the normal model understated tail losses by ~59% (ES_1% Empirical -5.0% vs. Normal -3.1%)
- 3% tail exceedances occur 2.8 times more often than normal predicted, consistent with high excess kurtosis (~13.6).
- Rolling Sharpe ratios remain relatively stable during calm periods but show meaningful deterioration during volatility spikes, reflecting clustered downside risk (eg. 2020 Covid-19 Shock).
- Tail-adjusted Sharpe (-4.11) reveals far more downside risk than static Sharpe (0.05), highlighting how volatility-based metrics understate the severity of fat-tail losses.

4. Tail-Risk Analysis

This section examines the downside tail of the SPY return distribution to evaluate how accurately Gaussian assumptions capture extreme or clustered losses.

As shown in the *key findings*, the distribution exhibits a mild negative skew and high excess kurtosis, indicating materially heavier left tails than a normal distribution predicts. This results in a leptokurtic return distribution, as illustrated in Figure 1. The lower 5% of returns contain losses significantly larger than Gaussian estimates, once more reflecting the negative skew and pronounced kurtosis.

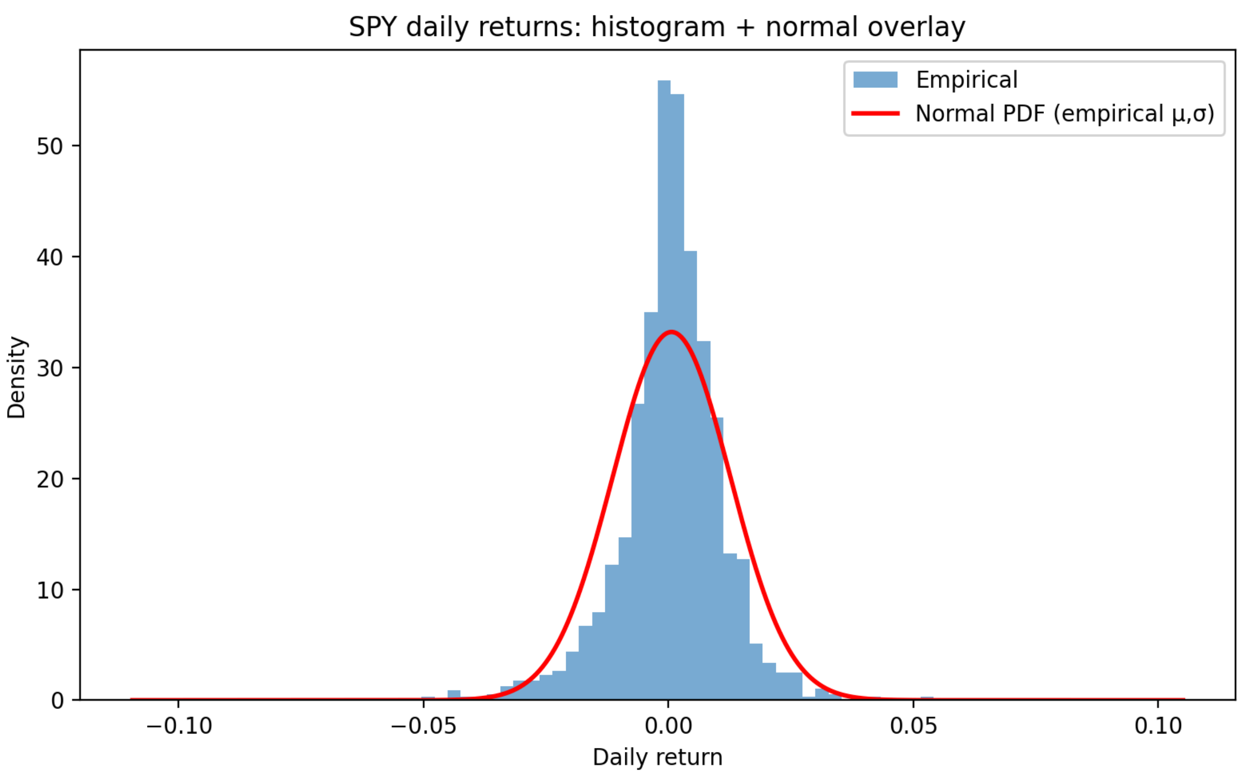

Figure 1. *Histogram of SPY daily returns with normal overlay. The empirical distribution shows negative skew and pronounced excess kurtosis relative to the Gaussian benchmark.*

Figure 2 further highlights the deviation from normality. The QQ plot shows the empirical quantiles diverging sharply from the 45º line in the lower tail, providing additional evidence of heavier expected downside extremes.

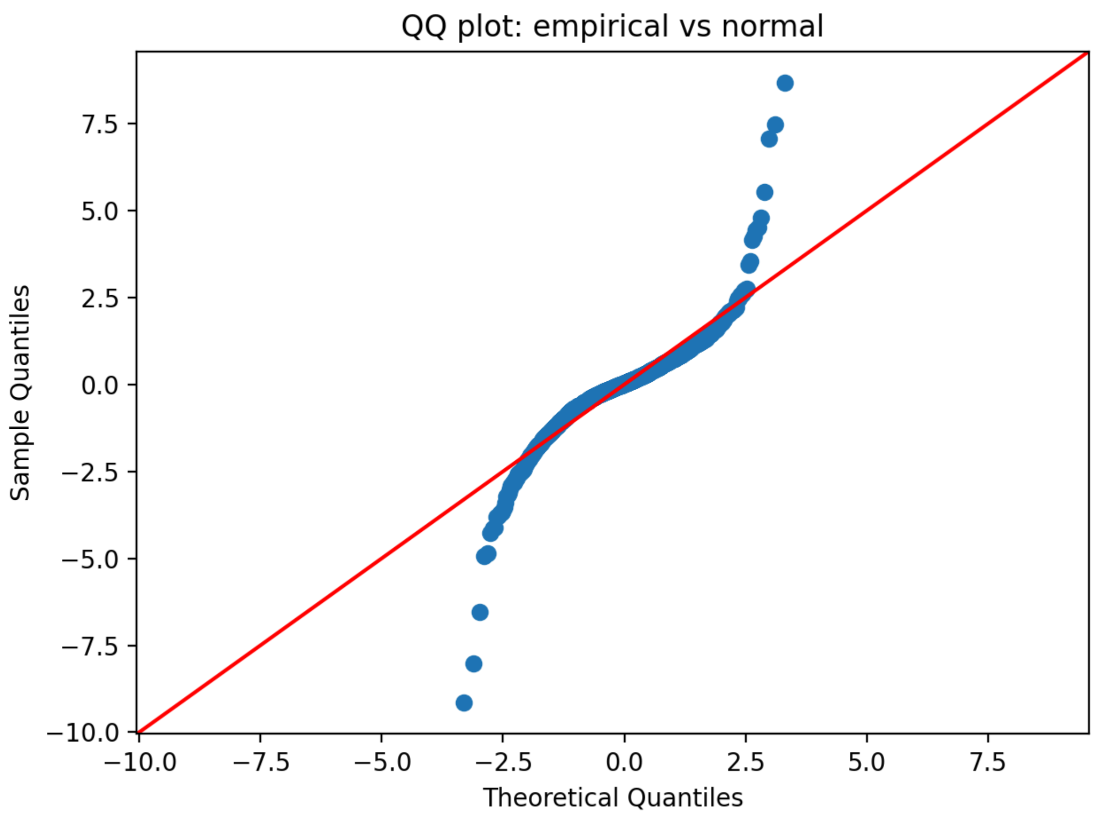

Figure 2. *QQ plot of SPY daily returns. The lower-tail deviation from the 45º line indicates heavier than normal downside extremes.*

Empirical losses during the worst 1% of days were approximately -5.0%, compared to only -3.1% under a normal model; this represents ~59% understatement during the most damaging tail-loss periods. Tail exceedances also occur more frequently than Gaussian assumptions predicted, with 3% tail exceedances occurring 2.8 times as often as implied. Figure 3 visualizes this discrepancy, showing the exceedance curves diverging sharply from each other.

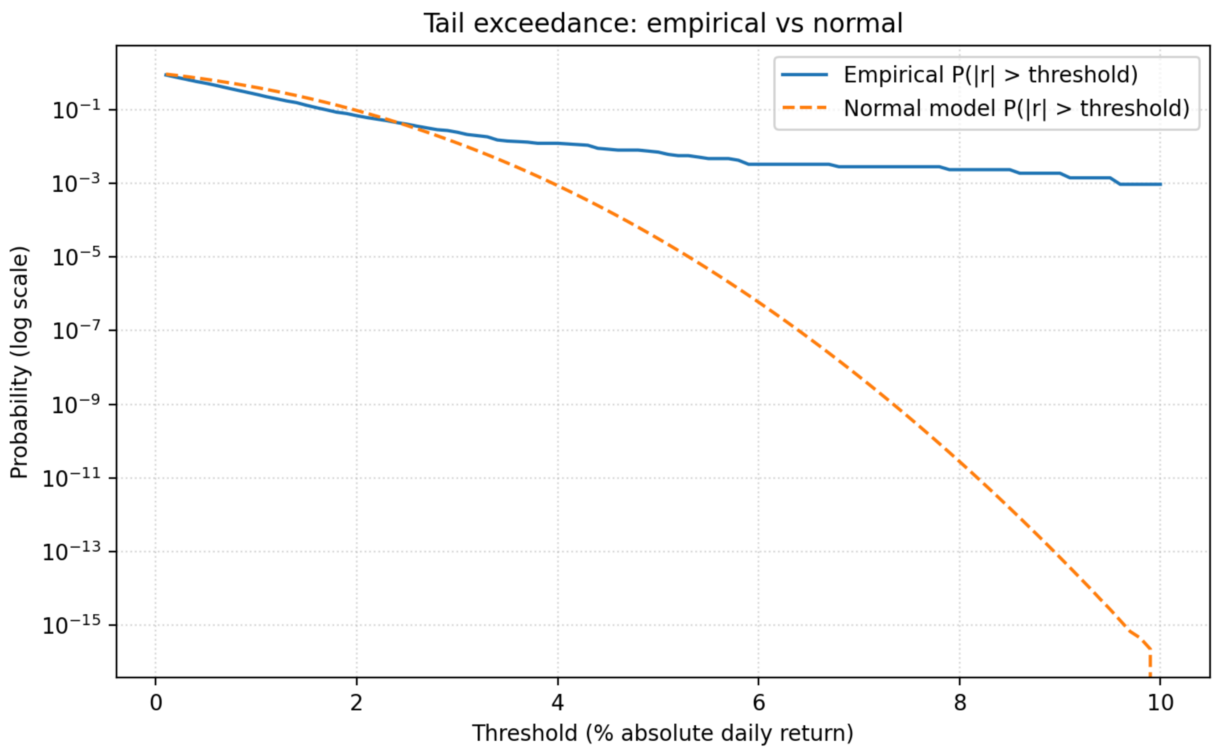

Figure 3. *Tail exceedance frequency for SPY daily returns. Empirical exceedances occur 2.8 times more often than Gaussian predictions, consistent with high excess kurtosis.*

This disconnect between Empirical returns and normal-model predictions highlights the limits of Gaussian-based risk frameworks, which assume symmetric returns and thin tails. In practice, negative skew and high excess kurtosis in SPY returns caused by clustered volatility and disproportionately large downside moves cause traditional volatility metrics to understate risk. Moreover, the normal model's understatement of tail size directly affects risk-adjusted performance metrics such as the Sharpe ratio, which treats upside and downside symmetrically. This next section evaluates how rolling and tail-adjusted Sharpe ratios attempt to cope with these empirical misalignments.

5. Dynamic Sharpe Ratios

Traditionally, the Sharpe ratio is computed as a single static value over a full sample period, implicitly assuming stable volatility and symmetric return behavior. In practice, real world SPY returns exhibit neither of these characteristics, instead showing clustered volatility and heavy downside tails. As a result, a traditional static Sharpe ratio inaccurately measures risk-adjusted performance. To address this issue, the script used to form this analysis computes a dynamic Sharpe ratio using a rolling 252-day window, allowing both mean and volatility of returns to evolve over time.

The rolling Sharpe ratio remains relatively stable during low volatility market regimes but deteriorates quickly during volatility spikes, most notably the 2020 Covid-19 shock during the tested time period. This pattern reflects the asymmetric nature of SPY's risk profile being sharp spikes in volatility during downturns and simultaneously declining returns, these combined cause the Sharpe to collapse. Figure 4 plots the rolling Sharpe and rolling tail-adjusted Sharpe ratios on the same axis, showing how the tail-adjusted measure collapses far more severely during volatility spikes.

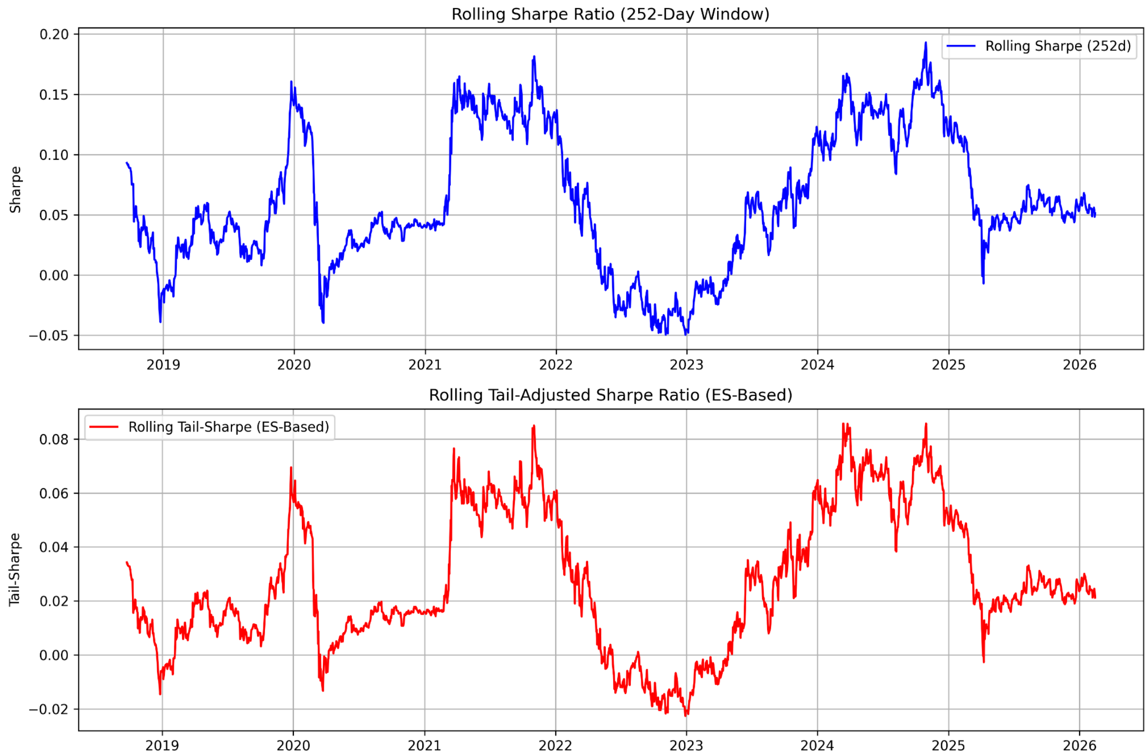

Figure 4. *Rolling Sharpe vs. rolling tail-adjusted Sharpe (252-day window). The two measures align during calm regimes but diverge sharply during downside volatility spikes, revealing the limitations of volatility-based Sharpe ratios.*

To more accurately capture downside risk, a tail-adjusted Sharpe ratio is also computed by replacing total volatility with a downside tail-risk measure. This metric highlights substantially weaker risk-adjusted performance (-4.11) compared with the static Sharpe ratio (0.05), revealing how volatility-based Sharpe measures systematically understate the severity of fat-tail losses. Together, the rolling and tail-adjusted Sharpe ratios illustrate how risk-adjusted performance varies across different market regimes and why Gaussian-based metrics fail to capture real-world downside risk.

6. Interpretations & Implications

The empirical results show a clear disconnect between Gaussian assumptions and the actual behavior of SPY's return distribution. While the normal model implies symmetric returns and thin tails, real-world data reveals a persistent negative skew, high excess kurtosis, and disproportionately large negative moves. These features produce larger and more frequent tail losses than Gaussian models estimate, ultimately causing traditional volatility-based metrics to underestimate true risk exposure.

The contrast between rolling Sharpe and tail-adjusted Sharpe ratios further illustrates how risk-adjusted performance varies across market regimes. During low volatility market regimes, the two measures appear nearly identical, giving the impression of stable returns. However, during volatility spikes, such as the 2020 Covid-19 shock, the tail-adjusted Sharpe more accurately reflects changes in market structure and collapses far more severely, capturing the concentration of losses in the left tail. Ultimately, this divergence shows that Sharpe ratios materially overstate risk-adjusted performance during low-volatility regimens preceding drawdown. This behavior reinforces the limitations of relying solely on static or volatility-based Sharpe ratios when evaluating assets with asymmetric or fat-tailed return distributions, such as SPY's leptokurtic behavior.

These findings carry important implications for portfolio construction and risk management. Models that assume market normality may understate downside risk, misprice tail events, and prioritize strategies that appear stable only during favorable, low volatility, market regimes. Incorporating tail-sensitive metrics, regime-aware assessments, and non-Gaussian modeling frameworks can help combat these issues, providing a more realistic view of the market especially during periods of high market stress or structural regime shifts.

7. Limitations

This analysis is subject to several limitations. First, it relies on historical SPY price action, which may not fully represent future market conditions. Second, the use of daily returns smooths intraday volatility and may understate extreme moves. Third, the rolling Sharpe ratio depends on the chosen window length, and different window sizes may yield different interpretations. Finally, the analysis does not account for transaction costs, slippage, or alternative risk measures that could refine the results.

8. Conclusion

Overall, the analysis highlights SPY's leptokurtic return distribution, its pronounced downside tail-risk characteristics, and the limitations of traditional volatility-based risk metrics. These results provide a foundation for understanding how real market behavior deviates from Gaussian assumptions and why Sharpe-style ratios often fail to accurately account for fat-tail losses.